

Kubernetes From Zero: install it **4 ways**

then break it, fix it, and clean up. A free hands-on lab.

THE USUAL TUTORIAL

\$ paste 20 commands → "it works" ... or not



kubeadm

2 Ubuntu VMs, built by hand



Minikube

a cluster in one container



MicroK8s

one snap, one command



Amazon EKS

managed, with real costs

install → verify → break → fix → compare → clean up

THE PAIN

Why first Kubernetes installs go wrong

the commands are easy to copy. Understanding is not.



Copy-paste installs

swap, br_netfilter, cgroups:
nobody says why



"NotReady" panic

a fresh control plane is
NotReady, and that is normal



Hidden cloud costs

a forgotten load balancer
or NAT keeps billing



No failure practice

the first real error
happens at work

THE IDEA

Learn to drive **four vehicles**

same road rules (Kubernetes), very different amounts of work under the hood



a car from parts

kubeadm

every piece visible

the simulator

Minikube

safe, on a laptop

a scooter

MicroK8s

small and quick

a taxi

Amazon EKS

the meter runs

and in every one, you stall it on purpose and learn why

WHO IT IS FOR

Use it when... skip it when...

a beginner lab, not a production blueprint

✅ USE IT WHEN YOU...

- 👤 have never installed Kubernetes
- 🎓 prepare for the CKA exam
- 🤖 must choose kubeadm vs Minikube vs EKS
- 🔧 want to debug NotReady nodes calmly
- 💡 want EKS without a surprise bill

⏏ SKIP IT WHEN YOU...

- need an HA production cluster (3 control planes, etcd backups)
- already run clusters and want GitOps or service meshes

HOW YOU USE IT

A roadmap in 10 levels

each level ends with something you can show



read the lesson → run it → compare with the real output

a guided tutorial walks you through every level

ARCHITECTURE

Same parts, different places

what each method installs, and where it puts it

CONTROL PLANE

-  kube-apiserver
-  etcd
-  scheduler
-  controller-manager

EVERY NODE

-  kubelet
-  containerd (CRI)
-  kube-proxy
-  CNI: Pod network

kubeadm

static Pods on your VM · Flannel · you own it

Minikube

a whole node inside one Docker container

MicroK8s

one process (kubelite) · dqlite · Calico

EKS

control plane run by AWS · VPC CNI · you run nodes

HOW IT WORKS

Every command is **really run**

the lessons are tested Markdown, not screenshots of someone's laptop



GitHub Actions, on every change

kubeadm (2-3 VMs) · Minikube · MicroK8s · install on Windows + macOS

Amazon EKS: run against a real AWS account

inventory before → create → test → delete

→ verify every resource is gone → inventory after

MEASURED RESULTS · LAB

The real EKS run, and real errors

numbers from the recorded run, eu-central-1

CLUSTER LIFETIME (minutes, LAB)

create 14.5

test + console

delete 11

≈ 41 min total · 868.9 s create · 674.9 s delete

\$0.15–0.20

list-price estimate

before = after

region inventory

35 s

minikube start in CI

ERRORS YOU REPRODUCE AND FIX

NetworkPluginNotReady: cni plugin not initialized

Kubelet stopped posting node status

0/1 nodes are available: Insufficient cpu

CrashLoopBackOff (a "forwardd" typo in CoreDNS)

connection to the server localhost:8080 was refused

STUDY MATERIAL

Everything to learn it properly

free, in one repository

13

concept lessons

4

install lessons

8

troubleshooting labs

5

challenges

15

capstone questions

12

tutorial chapters



18-chapter video · full + silent version



63-page study guide PDF



~70-term glossary · 25 interview questions

RUN IT YOURSELF

Six steps to your own cluster

Multipass VMs on Windows, macOS or Linux. No cloud needed.

```
1 $ git clone .../sufyan-devops-kubernetes-from-zero
```

```
2 $ multipass launch 24.04 --name k8s-cp ...
```

```
3 $ ... bash -s < kubeadm/scripts/prepare-node.sh
```

```
4 $ sudo kubeadm init --pod-network-cidr=...
```

```
5 $ kubectl apply -f .../kube-flannel.yml
```

```
6 $ sudo kubeadm join ... (on the worker)
```

k8s-cp Ready · k8s-worker Ready

each step is explained, with the real output to compare

FREE AND OPEN SOURCE

Install it. Break it. **Fix it.**

Which install method do you use at work, and why? 💬



The repository

[github.com/sufyanahmadkamboh/
sufyan-devops-kubernetes-from-zero](https://github.com/sufyanahmadkamboh/sufyan-devops-kubernetes-from-zero)



All my projects

sufyanahmadkamboh.github.io

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